The "Negative-Viscosity" Effect in a Magnetic Fluid: First-Principles Derivations of the Complete Theory

— SI (rationalized MKSA) formulation —

Bacri, Perzynski, Shliomis, Burde, PRL 75, 2128 (1995),
built up from its sources:
Shliomis, Sov. Phys. JETP 34, 1291 (1972) [Ref. 6];
Martsenyuk, Raikher, Shliomis, Sov. Phys. JETP 38, 413 (1974) [Ref. 8];
Shliomis & Morozov, Phys. Fluids 6, 2855 (1994) [Ref. 5];
Shliomis & Stepanov, JMMM 122, 176 (1993);
McTague, JCP 51, 133 (1969) [Ref. 7]

All source papers are written in Gaussian units. This document re-derives the theory in SI, where several equations change *form* (not merely numerical value) through factors of μ_0. Every boxed equation is re-derived and dimensionally verified; see Sec. 17.

Contents

1 What changes in SI, and why: the conversion rules **3**
1.1 The origin of every μ_0 . . . 3
1.2 Unit conversions used for the experimental numbers . . . 3
1.3 H-form vs. B-form . . . 3

2 Conventions, symbols, and the plan **4**

I Statistical mechanics of a rigid dipole **4**

3 Single-particle physics **4**
3.1 Rotational friction and the two microscopic times (no μ_0: purely mechanical) . . . 4
3.2 Néel vs. Brownian relaxation; the rigid-dipole regime . . . 5
3.3 Equilibrium averages (the μ_0 enters through U) . . . 5
3.4 SI recomputation of the paper's conversion factors . . . 5

4 The rotational Fokker–Planck equation **5**

5 The exact first-moment (magnetization) equation **6**

6 Three closures of the moment hierarchy **6**
6.1 Closure 1: near-isotropy ⇒ Debye (exact to $O(\xi)$) . . . 6
6.2 Closure 2: Shliomis (1972) . . . 6
6.3 Closure 3: effective field (MRSh) ⇒ PRL (5),(6) . . . 6

II Ferrohydrodynamics with internal rotation **7**

7 Coupled field–flow equations: the SI forms of PRL (1),(3),(4) 7
7.1 Kelvin force and magnetostatics . . . 7
7.2 Spin balance and the antisymmetric stress . . . 7
7.3 Eliminating spin inertia . . . 7

8 The rotational-viscosity theorem (contains *no* μ_0) 8

III The three field regimes 8

9 Static field: Shliomis' formula and the MRSh version 8

10 Weak alternating field: the sign change 9

11 Arbitrary amplitude, Debye closure; the corrected square-wave formula 9

IV The PRL's theory, numerics, and experiment 10

12 Perturbation theory in $\Omega\tau_B$: PRL (7)–(9) in SI 10

13 Numerical verification (dimensionless: identical to the Gaussian document) 10

14 The experiment in SI 11
14.1 Viscometry . . . 11
14.2 Data reduction, SI . . . 11
14.3 Findings . . . 11
14.4 Critical assessment (SI arithmetic) . . . 11

15 Zero total viscosity and the pump instability 12

16 Gaussian → SI dictionary (summary) 12

17 Verification ledger (SI-specific checks in bold) 13

18 Errata (carried over and verified in SI) 13

19 Open problems for follow-up work 13

1 What changes in SI, and why: the conversion rules

This is the part most SI conversions of ferrohydrodynamics get wrong, so it is stated explicitly before any physics.

1.1 The origin of every μ_0

In Gaussian units $\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$ and the energy of a moment is $U = -\mathbf{m}\cdot\mathbf{H}$. In SI,

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}), \qquad U = -\mathbf{m}\cdot\mathbf{B} = -\mu_0\,\mathbf{m}\cdot\mathbf{H}, \qquad \mathbf{T} = \mathbf{m}\times\mathbf{B} = \mu_0\,\mathbf{m}\times\mathbf{H}. \tag{1}$$

The magnetic moment couples to $\mathbf{B}$*, not to* $\mathbf{H}$. Every place where the Gaussian text writes $\mathbf{m}\times\mathbf{H}$, $\mathbf{M}\times\mathbf{H}$, mH/kT or $(\mathbf{M}\cdot\nabla)\mathbf{H}$, the SI expression carries one factor μ_0 (or, equivalently, is written with $\mathbf{B}$). Consequently:

Object	Gaussian	SI
Orientational energy	$-mH\cos\theta$	$-\mu_0 mH\cos\theta = -mB\cos\theta$
Langevin parameter	$\xi = mH/kT$	$\xi = \mu_0 mH/kT = mB/kT$
Torque density	$\mathbf{M}\times\mathbf{H}$	$\mu_0\,\mathbf{M}\times\mathbf{H} = \mathbf{M}\times\mathbf{B}$
Kelvin force density	$(\mathbf{M}\cdot\nabla)\mathbf{H}$	$\mu_0(\mathbf{M}\cdot\nabla)\mathbf{H}$
Initial susceptibility	$\chi = nm^2/3kT$	$\chi = \mu_0 nm^2/3kT$ (dimensionless)
Magnetostatics	$\nabla\cdot(\mathbf{H}+4\pi\mathbf{M}) = 0$	$\nabla\cdot(\mathbf{H}+\mathbf{M}) = 0$

Note the asymmetry that matters: ξ, $\Delta\eta/\eta$, φ, g, τ_B, $\Omega\tau_B$ are all *unchanged in value* — they are dimensionless or mechanical. The purely mechanical relations ($\tau_B = 3\eta V/kT$, $\tau_s = \rho_s d^2/60\eta$, $\Delta\eta = \frac{3}{2}\eta\varphi g$, the viscosity theorem) contain *no* μ_0 at all. Only the magnetic couplings acquire μ_0.

1.2 Unit conversions used for the experimental numbers

Quantity	Gaussian	SI
Field H	1 Oe	$10^3/4\pi = 79.577$ A/m
Flux density B	1 G	10^{-4} T
Magnetization M	1 G (i.e. $4\pi M$ in G)	10^3 A/m
Viscosity η	1 P = 1 g/(cm s)	0.1 Pa s; 1 cP=1 mPa s
Anisotropy K	1 erg/cm^3	0.1 J/m^3
Energy	1 erg	10^{-7} J
Moment m	1 erg/G	10^{-3} A m^2

The experiment's numbers therefore read, in SI:

$$\eta(0,0) = 77\,\text{mPa s}, \qquad \eta(220\,\text{mT},0) = 220\,\text{mPa s}, \qquad \eta_{\min} = 58\,\text{mPa s}, \tag{2}$$

$$\Delta\eta:\ +143\,\text{mPa s} \ \longrightarrow\ -19\,\text{mPa s}. \tag{3}$$

1.3 H-form vs. B-form

Two equivalent SI conventions exist. We use the **H-form** throughout (fields in A/m, explicit μ_0), because it maps one-to-one onto the source papers. The **B-form** is obtained everywhere by $\mu_0\mathbf{H}\to\mathbf{B}$, e.g. $\xi = mB/kT$, torque density $\mathbf{M}\times\mathbf{B}$; for the PRL geometry (long solenoid, nonmagnetic-limit sample) the two differ only in bookkeeping. Where a demagnetizing correction matters, $\mathbf{H}$ inside the sample is the correct variable, which is why the H-form is kept.

2 Conventions, symbols, and the plan

SI (rationalized MKSA) units throughout; $\mu_0 = 4\pi \times 10^{-7}$ H/m, $k_B = 1.380649 \times 10^{-23}$ J/K.

Symbol	SI unit	Meaning
η	Pa s	zero-field viscosity of the suspension (77 mPa s)
$\Delta\eta$	Pa s	rotational (vortex) viscosity contributed by the field
$\varphi = nV$	—	solid volume fraction (0.20); n = number density (m^{-3})
d, V, a	m,m^3,m	10 nm; $V = \pi d^3/6 = 5.24 \times 10^{-25}\,\mathrm{m}^3$; $a = d/2$
ρ_s	$\mathrm{kg/m}^3$	grain material density ($6000\,\mathrm{kg/m}^3$)
M_s	A/m	grain spontaneous magnetization, 400 G$= 4 \times 10^5$ A/m
$m = M_s V$	$\mathrm{A\,m}^2$	particle magnetic moment, $2.09 \times 10^{-19}\,\mathrm{A\,m}^2$
K	$\mathrm{J/m}^3$	anisotropy constant, $2 \times 10^6\,\mathrm{erg/cm}^3 = 2 \times 10^5\,\mathrm{J/m}^3$
$\sigma = KV/k_BT$	—	barrier parameter (≈ 26)
$\mathbf{H}$	A/m	applied field; $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$
$\xi = \mu_0 mH/k_BT$	—	Langevin parameter ($= mB_{\mathrm{app}}/k_BT$)
$L(\xi) = \coth\xi - 1/\xi$	—	Langevin function
$\mathbf{M}$	A/m	magnetization, $\mathbf{M} = nm\langle\mathbf{e}\rangle$
$\chi = \mu_0 nm^2/3k_BT$	—	initial susceptibility (dimensionless)
$\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \mathbf{v}$	s^{-1}	flow vorticity
$\boldsymbol{\Theta} \equiv \boldsymbol{\omega}_p$	s^{-1}	mean particle angular velocity
$\zeta_r = 6\eta V$	J s	rotational friction coefficient
$D_r = k_BT/\zeta_r$	s^{-1}	rotational diffusion coefficient
$\tau_B = 1/2D_r = 3\eta V/k_BT$	s	Brownian time (measured 1.6 ms)
τ_s, I	s,kg/m	spin-inertia time; moment of inertia per volume
$g(\xi_0, \omega\tau_B)$	—	normalized rotational viscosity, $\Delta\eta = \frac{3}{2}\eta\varphi g$

Plan. Part A (Secs. 3–6): statistical mechanics of one rigid dipole. Part B (Secs. 7–8): ferrohydrodynamics with internal rotation. Part C (Secs. 9–11): the three field regimes. Part D (Secs. 12–15): the PRL's theory, numerics, experiment. Sec. 17: verification ledger (including SI dimensional checks). Sec. 18: errata.

Part I
Statistical mechanics of a rigid dipole

3 Single-particle physics

3.1 Rotational friction and the two microscopic times (no μ_0: purely mechanical)

Low-Reynolds rotation of a sphere of radius a at $\boldsymbol{\omega}$ relative to the liquid:

$$\mathbf{T}_{\mathrm{fr}} = -8\pi\eta a^3\boldsymbol{\omega} = -6\eta V\boldsymbol{\omega} \quad \Rightarrow \quad \zeta_r = 6\eta V \ [\mathrm{J\,s}]. \tag{4}$$

Inertial time. $I_1\dot{\boldsymbol{\omega}} = -\zeta_r\boldsymbol{\omega}$ with $I_1 = \rho_s V d^2/10$:

$$\tau_s = \frac{I_1}{\zeta_r} = \frac{\rho_s d^2}{60\eta} = \frac{6000\,\mathrm{kg/m}^3 \times (10^{-8}\,\mathrm{m})^2}{60 \times 10^{-3}\,\mathrm{Pa\,s}} = 1.0 \times 10^{-11}\,\mathrm{s}. \tag{5}$$

Diffusional time. Einstein relation $D_r = k_BT/\zeta_r$; the $l = 1$ harmonic relaxes at $l(l+1)D_r = 2D_r$ (proved in Sec. 5):

$$\boxed{\tau_B = \frac{1}{2D_r} = \frac{3\eta V}{k_BT}} \qquad \left[\text{check: } \frac{\mathrm{Pa\,s}\cdot\mathrm{m}^3}{\mathrm{J}} = \frac{\mathrm{J\,s}}{\mathrm{J}} = \mathrm{s}\ \checkmark\right]. \tag{6}$$

Numerically, with $V = 5.24 \times 10^{-25}\,\mathrm{m}^3$, $k_BT = 3.99 \times 10^{-21}$ J ($T = 289$ K): $\tau_B = 3.94 \times 10^{-7}$ s for a bare grain in water ($\eta = 1$ mPa s), versus the *measured* 1.6 ms — the discrepancy analysed in Sec. 14. Since $\tau_s/\tau_B \sim 10^{-5}$, spin inertia is negligible throughout.

3.2 Néel vs. Brownian relaxation; the rigid-dipole regime

Parallel relaxation channels: $\tau^{-1} = \tau_N^{-1} + \tau_B^{-1}$ with $\tau_N \simeq \tau_0 e^{\sigma}$. For the Co-ferrite grains,

$$\sigma = \frac{KV}{k_B T} = \frac{2\times10^5\,\mathrm{J/m^3}\times 5.24\times10^{-25}\,\mathrm{m^3}}{3.99\times10^{-21}\,\mathrm{J}} \approx 26 \;\Rightarrow\; \tau_N \sim \tau_0 e^{26} \sim 10^2\,\mathrm{s} \gg \tau_B, \tag{7}$$

so the internal channel is frozen: *rigid dipole*, and any change of $\mathbf{M}$ requires mechanical rotation. The anisotropy field is where SI changes the form:

$$\boxed{H_a = \frac{2K}{\mu_0 M_s} = \frac{2\times2\times10^5\,\mathrm{J/m^3}}{4\pi\times10^{-7}\times4\times10^5\,\mathrm{A/m}} = 7.96\times10^5\,\mathrm{A/m}, \qquad \mu_0 H_a = \frac{2K}{M_s} = 1.00\,\mathrm{T}} \tag{8}$$

i.e. exactly the 10^4 Oe of the paper. Note $H_a = 2K/\mu_0 M_s$ in SI versus $H_a = 2K/M_s$ in Gaussian — a genuine formula change, and $\mu_0 H_a = 2K/M_s$ recovers the Gaussian-looking expression in tesla.

3.3 Equilibrium averages (the μ_0 enters through U)

$$U = -\mathbf{m}\cdot\mathbf{B}_{\mathrm{app}} = -\mu_0 m H\cos\theta = -k_B T\,\xi\cos\theta, \qquad \boxed{\xi = \frac{\mu_0 m H}{k_B T} = \frac{mB}{k_B T}} \tag{9}$$

$$Z(\xi) = \int e^{\xi\cos\theta}\mathrm{d}\Omega_{\mathbf{e}} = \frac{4\pi\sinh\xi}{\xi}, \qquad \langle\cos\theta\rangle_0 = \frac{\mathrm{d}\ln Z}{\mathrm{d}\xi} = \coth\xi - \frac{1}{\xi} = L(\xi), \tag{10}$$

$$\boxed{\mathbf{M}_0 = nm\,L(\xi)\,\mathbf{h} = \varphi M_s L(\xi)\,\mathbf{h}} \quad [\mathrm{A/m}], \qquad \boxed{\chi = \frac{\mu_0 n m^2}{3k_B T}\ \text{(dimensionless)}} \tag{11}$$

Dimensional check of χ: $[\mathrm{H/m}][\mathrm{m^{-3}}][\mathrm{A^2\,m^4}]/[\mathrm{J}] = \mathrm{H\,A^2/(m\,J)} = 1$ ✓ (using $\mathrm{H\,A^2{=}J}$). Numerically $\chi = 1.76$ for the PRL sample. Second moment (unchanged, since it is pure statistics):

$$L' = 1 - \coth^2\xi + \xi^{-2}, \qquad L^2 + L' = 1 - \frac{2L}{\xi} \;\Rightarrow\; \langle\cos^2\theta\rangle_0 = 1 - \frac{2L}{\xi}, \tag{12}$$

$$\boxed{\langle\mathbf{ee}\rangle_0 = \frac{L}{\xi}\mathbb{1} + \Big(1 - \frac{3L}{\xi}\Big)\hat{\boldsymbol{\nu}}\hat{\boldsymbol{\nu}}} \qquad (\mathrm{trace} = 1\ \checkmark;\ \xi\to0:\ \mathbb{1}/3;\ \xi\to\infty:\ \hat{\boldsymbol{\nu}}\hat{\boldsymbol{\nu}}). \tag{13}$$

3.4 SI recomputation of the paper's conversion factors

$$m = M_s V = 4\times10^5\,\mathrm{A/m}\times5.24\times10^{-25}\,\mathrm{m^3} = 2.09\times10^{-19}\,\mathrm{A\,m^2}, \tag{14}$$

$$\xi = \frac{\mu_0 m}{k_B T}H = \frac{4\pi\times10^{-7}\times2.09\times10^{-19}}{3.99\times10^{-21}}\,H = 6.60\times10^{-5}\,H\,[\mathrm{A/m}] = 0.0525\,B\,[\mathrm{mT}], \tag{15}$$

and since $1\,\mathrm{Oe} = 79.58\,\mathrm{A/m}$: $\xi = 5.25\times10^{-3}H[\mathrm{Oe}]$ — *exactly* the paper's coefficient (our SI computation gives 5.249×10^{-3} at $T = 289\,\mathrm{K}$). Likewise $\omega\tau_B = 10^{-2} f[\mathrm{Hz}]$ gives $\tau_B = 1.59\,\mathrm{ms}$. Useful SI forms:

$$\xi_0 = 0.0660\,H_0\,[\mathrm{kA/m}] = 0.0525\,B_0\,[\mathrm{mT}], \qquad 2000\,\mathrm{Oe} = 159\,\mathrm{kA/m} = 200\,\mathrm{mT} \;\Rightarrow\; \xi_0 = 10.5. \tag{16}$$

4 The rotational Fokker–Planck equation

Torque balance on one particle (inertia dropped), *with the SI magnetic torque*:

$$0 = \mu_0\,\mathbf{m}\times\mathbf{H} - \zeta_r(\boldsymbol{\omega} - \boldsymbol{\Omega}) + \mathbf{T}_{\mathrm{th}} \;\Rightarrow\; \boldsymbol{\omega}_{\mathrm{det}} = \boldsymbol{\Omega} + \frac{\mu_0}{\zeta_r}\,\mathbf{m}\times\mathbf{H}. \tag{17}$$

With $\hat{\mathbf{R}} = \mathbf{e}\times\partial/\partial\mathbf{e}$ and $U = -\mu_0\mathbf{m}\cdot\mathbf{H}$:

$$\mu_0\,\mathbf{m}\times\mathbf{H} = -\hat{\mathbf{R}}U \qquad [\text{check: } \hat{\mathbf{R}}U = -\mu_0 m H\,\mathbf{e}\times\mathbf{h} = -\mu_0\mathbf{m}\times\mathbf{H}\ \checkmark], \tag{18}$$

so the Fokker–Planck (rotational Smoluchowski) equation retains its Gaussian *form*, the μ_0 hiding inside U and ξ:

$$\boxed{\frac{\partial W}{\partial t} = \hat{\mathbf{R}}\cdot\Big[D_r\Big(\hat{\mathbf{R}}W + \frac{W}{k_BT}\hat{\mathbf{R}}U\Big) - \boldsymbol{\Omega}W\Big]}, \qquad \frac{U}{k_BT} = -\xi\,(\mathbf{e}\cdot\mathbf{h}). \tag{19}$$

Checks: Gibbs state annihilates the bracket ($\hat{\mathbf{R}}W_0 = -W_0\hat{\mathbf{R}}U/k_BT$) ✓; norm conserved ✓; isotropic W stationary under pure $\boldsymbol{\Omega}$ ✓.

Operator identities (unchanged, pure geometry).

$$\hat{\mathbf{R}}_j e_i = \varepsilon_{jki}e_k, \qquad \hat{\mathbf{R}}^2 e_i = -2e_i, \qquad (\hat{\mathbf{R}}e_i)\cdot(\mathbf{e}\times\mathbf{h}) = h_i - e_i(\mathbf{e}\cdot\mathbf{h}). \tag{20}$$

5 The exact first-moment (magnetization) equation

Multiplying (19) by e_i and integrating by parts, term by term as in the Gaussian version (diffusion $\to -2D_r\langle e_i\rangle$; potential drift $\to \xi D_r\langle h_i - e_i(\mathbf{eh})\rangle$ using $\hat{\mathbf{R}}U/k_BT = -\xi\,\mathbf{e}\times\mathbf{h}$; convection $\to (\boldsymbol{\Omega}\times\langle\mathbf{e}\rangle)_i$), and using $2D_r = 1/\tau_B$:

$$\boxed{\frac{\mathrm{d}\langle\mathbf{e}\rangle}{\mathrm{d}t} = \boldsymbol{\Omega}\times\langle\mathbf{e}\rangle - \frac{1}{\tau_B}\langle\mathbf{e}\rangle + \frac{\xi}{2\tau_B}\big(\mathbf{h} - \langle\mathbf{ee}\rangle\cdot\mathbf{h}\big)} \tag{21}$$

Exact but unclosed. *Form identical to Gaussian* — because the entire μ_0 dependence was absorbed into the definition $\xi = \mu_0 mH/k_BT$. This is the single most useful structural fact of the SI conversion: *all* kinetic equations for $\langle\mathbf{e}\rangle$ keep their form; only the *dictionary* $\xi \leftrightarrow H$ and the torque-to-hydrodynamics couplings change.

Equilibrium verification. $\langle\mathbf{e}\rangle = L\mathbf{h}$, $\langle\mathbf{ee}\rangle\cdot\mathbf{h} = (1 - 2L/\xi)\mathbf{h}$ give RHS $= -\frac{L}{\tau_B}\mathbf{h} + \frac{\xi}{2\tau_B}\frac{2L}{\xi}\mathbf{h} = 0$ ✓.

6 Three closures of the moment hierarchy

6.1 Closure 1: near-isotropy ⇒ Debye (exact to $O(\xi)$)

$\langle\mathbf{ee}\rangle = \mathbb{1}/3 + O(\xi)$ turns the last term of (21) into $\xi\mathbf{h}/3\tau_B$; with $\mathbf{M} = nm\langle\mathbf{e}\rangle$ and the SI susceptibility $\chi = \mu_0 nm^2/3k_BT$,

$$\boxed{\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\mathbf{M} - \frac{1}{\tau_B}\big(\mathbf{M} - \chi\mathbf{H}(t)\big)} \qquad (\xi \ll 1,\ \text{any } \omega). \tag{22}$$

Form unchanged; the μ_0 sits inside χ. (Equivalently $\mathbf{M}_0 = \chi\mathbf{B}/\mu_0$.) The Debye law is a theorem, not a postulate.

6.2 Closure 2: Shliomis (1972)

$$\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \boldsymbol{\Theta}\times\mathbf{M} - \frac{1}{\tau_B}\big(\mathbf{M} - \mathbf{M}_0(t)\big), \qquad \mathbf{M}_0 = \varphi M_s L(\xi(t))\mathbf{h}. \tag{SM 7}$$

Exact for (a) $\xi \ll 1$, (b) static $\mathbf{H}$ of any strength; uncontrolled for finite ξ *and* $\omega\tau_B$.

6.3 Closure 3: effective field (MRSh) ⇒ PRL (5),(6)

Ansatz $W_e \propto \exp(\boldsymbol{\xi}_e\cdot\mathbf{e})$, so

$$\boxed{\mathbf{M} = \varphi M_s L(\xi_e)\frac{\boldsymbol{\xi}_e}{\xi_e}} \tag{PRL 5}$$

and $\langle \mathbf{ee} \rangle_e$ from (13) with $\hat{\boldsymbol{\nu}} = \hat{\boldsymbol{\xi}}_e$. Substituting into (21) and regrouping (the coefficient-by-coefficient proof is identical to the Gaussian document: expand $-\frac{L_e}{\tau_B \xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi})$ and $\boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi}) = \xi_e^2 \xi [(\hat{\boldsymbol{\xi}}_e \mathbf{h}) \hat{\boldsymbol{\xi}}_e - \mathbf{h}]$, then match the $\mathbf{h}$ and $\hat{\boldsymbol{\xi}}_e$ projections):

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\Big(L_e \frac{\boldsymbol{\xi}_e}{\xi_e}\Big) = \boldsymbol{\Omega} \times \Big(L_e \frac{\boldsymbol{\xi}_e}{\xi_e}\Big) - \frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi}) - \frac{1}{2\tau_B \xi_e^2}\Big(1 - \frac{3L_e}{\xi_e}\Big)\boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi})} \qquad \text{(PRL 6)}$$

with $\boldsymbol{\xi} = \xi \mathbf{h}$, $\xi = \mu_0 m H / k_B T$ — **the only SI modification is inside $\boldsymbol{\xi}$.** Linearized times (unchanged):

$$\tau_{\parallel} = \tau_B \frac{\mathrm{d}\ln L}{\mathrm{d}\ln \xi}, \qquad \tau_{\perp} = \frac{2\tau_B \xi}{\xi - L} \qquad (\text{both} \to \tau_B \text{ at } \xi \to 0;\ \to \tau_B/\xi,\ 2\tau_B/\xi \text{ at } \xi \gg 1). \qquad (23)$$

The Debye closure has the single time τ_B — the reason it fails at finite $\xi, \omega\tau_B$.

Part II
Ferrohydrodynamics with internal rotation

7 Coupled field–flow equations: the SI forms of PRL (1),(3),(4)

7.1 Kelvin force and magnetostatics

$$\mathbf{F}_{\mathrm{Kelvin}} = \mu_0 (\mathbf{M}\cdot\nabla)\mathbf{H}\ [\mathrm{N/m^3}], \qquad \nabla\cdot\mathbf{v} = 0, \quad \nabla \times \mathbf{H} = 0, \quad \nabla\cdot(\mathbf{H} + \mathbf{M}) = 0. \qquad (24)$$

(Both the μ_0 in the force and the disappearance of 4π from the divergence relation are SI changes.) In the PRL geometry $\mathbf{H}$ is uniform, so this force vanishes; it is kept for completeness.

7.2 Spin balance and the antisymmetric stress

$$I\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \mu_0\, \mathbf{M} \times \mathbf{H} - \frac{I}{\tau_s}(\boldsymbol{\Theta} - \boldsymbol{\Omega}), \qquad \frac{I}{\tau_s} = 6\eta\varphi\ [\mathrm{Pa\,s}], \qquad \text{(SM 8)}$$

where $I = \varphi \rho_s d^2/10$ [kg/m] and $\tau_s = \rho_s d^2/60\eta$ give $I/\tau_s = 6\eta\varphi$ ✓ (purely mechanical: no μ_0). Torque density transmitted to the fluid ⇒ antisymmetric stress $\sigma^a_{ik} = -\frac{1}{2}\varepsilon_{ikl} A_l$, $f_i = \frac{1}{2}(\nabla \times \mathbf{A})_i$; total angular-momentum conservation verifies the factor and sign. Hence

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + \mu_0 (\mathbf{M}\cdot\nabla)\mathbf{H} + \eta\Delta\mathbf{v} + \frac{I}{2\tau_s}\nabla \times (\boldsymbol{\Theta} - \boldsymbol{\Omega}). \qquad \text{(SM 6)}$$

7.3 Eliminating spin inertia

$\tau_s/\tau_B \sim 10^{-5} \Rightarrow$ drop $I\dot{\boldsymbol{\Theta}}$:

$$\boxed{6\eta\varphi\,(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \mu_0\, \mathbf{M} \times \mathbf{H} = \mathbf{M} \times \mathbf{B}} \qquad \text{(PRL 1, SI)}$$

Dimensional check: LHS $= \mathrm{Pa\,s} \cdot \mathrm{s^{-1}} = \mathrm{Pa}$; RHS $= \mathrm{H/m} \cdot \mathrm{A/m} \cdot \mathrm{A/m} = \mathrm{N/m^2} = \mathrm{Pa}$ ✓ (torque density has units of pressure). Substituting into (SM 6), with $\frac{I}{2\tau_s} \cdot \frac{1}{6\eta\varphi} = \frac{1}{2}$:

$$\boxed{\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + \mu_0 (\mathbf{M}\cdot\nabla)\mathbf{H} + \eta\Delta\mathbf{v} + \frac{\mu_0}{2}\nabla \times (\mathbf{M} \times \mathbf{H})} \qquad \text{(PRL 3, SI)}$$

(the PRL misprints $\nabla\rho$ for ∇p), and into (SM 7), using $(\mathbf{M} \times \mathbf{H}) \times \mathbf{M} = -\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$:

$$\boxed{\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \mathbf{M} - \frac{1}{\tau_B}(\mathbf{M} - \mathbf{M}_0) - \frac{\mu_0}{6\eta\varphi}\,\mathbf{M} \times (\mathbf{M} \times \mathbf{H})} \qquad \text{(PRL 4, SI)}$$

Check of the last term: $\mathrm{H/m} \cdot (\mathrm{A/m})^3/\mathrm{Pa\,s} = \mathrm{A/(m\,s)}$ ✓ (same as $\mathrm{d}M/\mathrm{d}t$).

8 The rotational-viscosity theorem (contains *no* μ_0)

Define g by the period-averaged torque density

$$\mu_0 \overline{\mathbf{M} \times \mathbf{H}} = -6\eta\varphi\, g\, \mathbf{\Omega}. \tag{25}$$

Proof 1. Incompressibility gives $\Delta\mathbf{v} = -2\nabla \times \mathbf{\Omega}$, so

$$\eta\Delta\mathbf{v} + \frac{\mu_0}{2}\nabla \times \overline{\mathbf{M} \times \mathbf{H}} = -2\eta\Big(1 + \tfrac{3}{2}\varphi g\Big)\nabla \times \mathbf{\Omega}, \tag{PRL 10}$$

i.e. Navier–Stokes with $\eta + \Delta\eta$ and

$$\boxed{\Delta\eta = \tfrac{3}{2}\,\eta\varphi\, g = \frac{3}{2}\eta\varphi\,\frac{\Omega - \omega_p}{\Omega}} \tag{PRL 2}$$

using $\boldsymbol{\omega}_p = (1-g)\mathbf{\Omega}$ from (PRL 1, SI) and (25). **Proof 2:** average (PRL 1, SI) and compare. Both agree ✓. *Note:* the theorem, its prefactor $\frac{3}{2}$, and the DC bound $0 \le g \le 1$ (hence $0 \le \Delta\eta \le \frac{3}{2}\eta\varphi$ for static fields) are unit-system independent, as they must be — g and $\Delta\eta/\eta$ are dimensionless ratios of mechanical quantities.

Part III
The three field regimes

9 Static field: Shliomis' formula and the MRSh version

Geometry $\mathbf{H} \perp \mathbf{\Omega}$ ($\beta = \pi/2$), first order in $\Omega\tau_B \ll 1$. Steady state of (SM 7) gives $M_y = \omega_p\tau_B M_x$, $M_x \simeq M_0$, torque density $\mu_0(\mathbf{M} \times \mathbf{H})_z = -\mu_0\omega_p\tau_B M_0 H$. The key SI identity (verified numerically to machine precision):

$$\boxed{\mu_0\,\tau_B M_0 H = \frac{3\eta V}{k_B T}\cdot\frac{\varphi}{V}\,mL(\xi)\,\mu_0 H = 3\eta\varphi\,\xi L(\xi)} \tag{26}$$

— note how $\mu_0 H$ combines with m/k_BT to reproduce exactly ξ. Inserting into (PRL 1, SI):

$$6\eta\varphi(\omega_p - \Omega) = -3\eta\varphi\,\xi L\,\omega_p \;\Rightarrow\; \omega_p = \frac{\Omega}{1 + \frac{1}{2}\xi L} \;\Rightarrow\; \boxed{\Delta\eta = \frac{3}{2}\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta} \tag{SM 2}$$

Limits: $\Delta\eta = \frac{1}{4}\eta\varphi\xi^2$ ($\xi \ll 1$); $\to \frac{3}{2}\eta\varphi$ ($\xi \to \infty$, $\omega_p = 0$, $g = 1$). MRSh version (same algebra with (PRL 6)):

$$\boxed{\Delta\eta^{\mathrm{MRSh}} = \frac{3}{2}\eta\varphi\,\frac{\xi L^2(\xi)}{\xi - L(\xi)}} \tag{27}$$

agreeing at both limits, but differing by up to 16% at $\xi \sim 3$–5:

ξ	0.5	1	2	3	5	10
g^{MRSh}	0.0400	0.1426	0.3948	0.5812	0.7621	0.8901
g^{Sh72}	0.0394	0.1353	0.3495	0.5019	0.6667	0.8182

10 Weak alternating field: the sign change

$\mathbf{H} = H_0 \cos\omega t\, \hat{\mathbf{x}}$, $\xi_0 = \mu_0 m H_0/k_B T \ll 1$ (Debye exact here). Phasor solutions:

$$M_x = \chi H_0 \cos\alpha \cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B, \tag{SM 30}$$

$$M_y = \Omega\tau_B \chi H_0 \cos^2\alpha \cos(\omega t - 2\alpha). \tag{SM 23,32}$$

Mean torque density, with the SI μ_0:

$$\mu_0 \overline{(\mathbf{M}\times\mathbf{H})_z} = -\tfrac{1}{2}\mu_0 \Omega\tau_B \chi H_0^2 \cos^2\alpha \cos 2\alpha, \tag{28}$$

and the second key SI identity (numerically verified exact):

$$\boxed{\mu_0\, \tau_B \chi H_0^2 = \frac{3\eta V}{k_B T}\cdot\frac{\mu_0 n m^2}{3k_B T}\cdot\mu_0 H_0^2 = \eta\varphi\,\xi_0^2} \tag{29}$$

(one μ_0 from the torque, one inside χ; together they build $\xi_0^2 = (\mu_0 m H_0/k_B T)^2$). Hence

$$g = \frac{\xi_0^2}{12}\cos^2\alpha\cos 2\alpha, \qquad \boxed{\Delta\eta = \frac{1}{8}\eta\varphi\xi_0^2\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2} = \frac{1}{4}\eta\varphi\bar{\xi}^2\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}} \tag{SM 28}$$

$\bar{\xi} = \xi_0/\sqrt{2}$ (SM's rms convention). Quasi-static check: $\omega \to 0$ must give $\overline{\frac{1}{4}\eta\varphi\xi_0^2\cos^2\omega t} = \frac{1}{8}\eta\varphi\xi_0^2$ ✓.

All qualitative results (unchanged; $R(x) = (1-x^2)/(1+x^2)^2$).

1. Sign change at $\omega_0\tau_B = 1$, i.e. $f_0 = 1/2\pi\tau_B = 100\,\mathrm{Hz}$ for $\tau_B = 1.59\,\mathrm{ms}$.
2. $R'(x) = 2x(x^2-3)/(1+x^2)^3 = 0 \Rightarrow x = \sqrt{3}$; $R(\sqrt{3}) = -1/8$: $\Delta\eta_{\min} = -\Delta\eta(0)/8$ at $\omega_m\tau_B = \sqrt{3}$.
3. High-frequency tail — *here SI changes the closed form.* With $\bar{\xi}/\tau_B = \mu_0 M_s \bar{H}/3\eta$ (from $m = M_s V$, $\tau_B = 3\eta V/k_B T$):

$$\boxed{\Delta\eta \simeq -\frac{\eta\varphi\bar{\xi}^2}{4\omega^2\tau_B^2} = -\frac{\varphi}{\eta}\Big(\frac{\mu_0 M_s \bar{H}}{6\omega}\Big)^2 = -\frac{\varphi}{\eta}\Big(\frac{M_s \bar{B}}{6\omega}\Big)^2} \tag{SM 29, SI}$$

   Check: $(\mathrm{A/m})^2(\mathrm{T})^2/(\mathrm{Pa\,s}\cdot\mathrm{s}^{-2})$; $\mathrm{A/m}\cdot\mathrm{T} = \mathrm{Pa}$, so $= \mathrm{Pa}^2/(\mathrm{Pa\,s}\cdot\mathrm{s}^{-2})\cdot\mathrm{s}^{-2}\ldots$ collecting: $\mathrm{Pa}^2\mathrm{s}^2/(\mathrm{Pa\,s}) = \mathrm{Pa\,s}$ ✓.
4. Mean spin $\Theta = \Omega[1 - \frac{\bar{\xi}^2}{6}R(\omega\tau_B)]$: $\omega\tau_B < 1 \Rightarrow \Theta < \Omega$ (drag); $\omega\tau_B > 1 \Rightarrow \Theta > \Omega$ (spin-up, energy field→particles→flow).

11 Arbitrary amplitude, Debye closure; the corrected square-wave formula

With $\xi(t) = \xi\gamma(t)$, $M_x = nmF$, $M_y = nm\Omega\tau_B G$:

$$\tau_B\dot{F} = L[\xi\gamma(t)] - F, \qquad \tau_B\dot{G} = F - \big(1 + \tfrac{1}{2}\xi\gamma F\big)G, \tag{SM 36}$$

$$\mu_0\overline{(\mathbf{M}\times\mathbf{H})_z} = -3\eta\varphi\,\xi\,\Omega\,\overline{G\gamma} \;\Rightarrow\; g = \tfrac{1}{2}\xi\overline{G\gamma}, \quad \Delta\eta = \tfrac{3}{4}\eta\varphi\,\xi\overline{G\gamma}, \tag{SM 37,38}$$

using $\mu_0 nmH\tau_B = 3\eta\varphi\xi$ [same structure as (26)]. Averaging (PRL 1, SI) shows $\frac{1}{2}\xi\overline{G\gamma} = (\Omega - \overline{\Theta})/\Omega$, i.e. (PRL 2) is exact at all amplitudes within this closure.

Square wave, weak field. Linearize ($L \to \xi\gamma/3$), define $\varepsilon = \pi/2\omega\tau_B$, $E = e^{-2\varepsilon}$; antiperiodicity $F(t+T/2) = -F(t)$ gives $F_0 = -\frac{\xi}{3}\tanh\varepsilon$; the resonant solution for G plus antiperiodicity gives $C = -(2a + 2\varepsilon bE)/(1+E)$ with $a = \xi/3$, $b = -a(1+\tanh\varepsilon)$; the average collapses to

$$\boxed{\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\Big(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\Big), \qquad \varepsilon = \frac{\pi}{2\omega\tau_B}} \qquad \text{(SM 40, corrected)}$$

Checks: $\varepsilon \to \infty \Rightarrow \frac{1}{4}\eta\varphi\xi^2$ ✓; $\varepsilon \to 0 \Rightarrow -\varepsilon^2/3 < 0$ ✓; numerically $\varepsilon_0 = 1.6061 \Rightarrow \omega_0\tau_B = 0.978$, $\varepsilon_m = 0.9199 \Rightarrow \omega_m\tau_B = 1.7075$, depth $-0.1050\Delta\eta(0)$ — reproducing SM's quoted 0.98, 1.707, −0.105 exactly. The scanned SM formula (missing the square on tanh) fails all three; see Sec. 18.

Part IV
The PRL's theory, numerics, and experiment

12 Perturbation theory in $\Omega\tau_B$: PRL (7)–(9) in SI

$\Omega\tau_B \lesssim 0.1$ ($\tau_B = 1.6\,\text{ms}$, $\Omega \lesssim 60\,\text{s}^{-1}$). Expand $\boldsymbol{\xi}_e = \xi_e(t)\mathbf{h} + \boldsymbol{\delta}$, $\boldsymbol{\xi} = \xi_0\cos\omega t\,\mathbf{h}$ with $\xi_0 = \mu_0 m H_0/k_BT$.

Zeroth order (project (PRL 6) on $\mathbf{h}$):

$$\boxed{\tau_B\frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\Big(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\Big)^{-1}\Big(1 - \frac{\xi_0}{\xi_e}\cos\omega t\Big)}, \qquad \Big(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\Big)^{-1} = \frac{L_e}{L'(\xi_e)}. \qquad \text{(PRL 7)}$$

First order: with $\mathbf{M}^{(1)} = \varphi M_s L(\xi_e) F\tau_B\,\boldsymbol{\Omega}\times\mathbf{h}$, transverse projection, then eliminating $\dot{L}_e$ via (PRL 7):

$$\boxed{\tau_B\frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \frac{1}{2}\Big(\frac{1}{L_e} - \frac{1}{\xi_e}\Big)\xi_0 F\cos\omega t} \qquad \text{(PRL 8)}$$

Torque, now with the SI factor and the third key identity $\mu_0\varphi M_s H_0\tau_B = 3\eta\varphi\xi_0$:

$$\mu_0\,\mathbf{M}^{(1)}\times\mathbf{H} = -3\eta\varphi\,\xi_0\cos\omega t\,L(\xi_e)F\,\boldsymbol{\Omega} \;\Rightarrow\; \boxed{g = \tfrac{1}{2}\xi_0\overline{\cos\omega t\;L(\xi_e)F}, \quad \Delta\eta = \tfrac{3}{2}\eta\varphi g} \qquad \text{(PRL 9)}$$

Checks: static limit $\dot{F} = 0 \Rightarrow F = 2L/(\xi - L)$, $g = \xi L^2/(\xi - L)$ = MRSh static formula ✓; weak field reproduces (SM 28) ✓.

Numerically regular form. $1/L_e$ diverges as $\xi_e \to 0$; integrate $P \equiv L(\xi_e)F$ instead:

$$\tau_B\dot{P} = L(\xi_e) - P - \tfrac{1}{2}P\,s(\xi_e)\,\xi_0\cos\omega t, \qquad s(\xi) = \frac{1}{L(\xi)} - \frac{3}{\xi} = \frac{\xi}{5} + O(\xi^3), \qquad g = \tfrac{1}{2}\xi_0\overline{\cos\omega t\,P}. \quad (30)$$

13 Numerical verification (dimensionless: identical to the Gaussian document)

Since g depends only on $(\xi_0, \omega\tau_B)$, the numerics are unit-system independent. RK4 integration of (ξ_e, P), ≥ 1200 steps/period, ≥ 20 periods:

$g(\xi_0, \omega\tau_B)$	ξ_0=4	8	12	16	20
$\omega\tau_B = 0.5$	0.295	0.451	0.517	0.554	0.580
$\omega\tau_B = 1.0$	0.047	0.163	0.231	0.275	0.306
$\omega\tau_B = 1.5$	−0.105	−0.110	−0.044	0.009	0.049
$\omega\tau_B = 2.0$	−0.140	−0.312	−0.299	−0.246	−0.197

matching PRL Fig. 3 (minimum ≈ -0.31 at $\omega\tau_B = 2$, $\xi_0 \sim 9$; zero crossings). Weak-field check at $\xi_0 = 0.3$: $g = \xi_0^2 R(\omega\tau_B)/12$ to $< 1\%$, exact zero at $\omega\tau_B = 1$. In SI field variables the same table is read with

$$\xi_0 = 0.0525\, B_0[\mathrm{mT}] \quad \Rightarrow \quad \xi_0 = 4, 8, 12, 16, 20 \;\longleftrightarrow\; B_0 = 76, 152, 229, 305, 381 \text{ mT}. \tag{31}$$

14 The experiment in SI

14.1 Viscometry

Horizontal capillary $d_c = 1\,\mathrm{mm}$ coaxial in a solenoid ($\mathbf{H} \parallel$ flow $\Rightarrow \mathbf{H} \perp \mathbf{\Omega}$, $\beta = \pi/2$); hydrostatic head h; for a Newtonian fluid

$$Q = \frac{\pi \rho^2 g_{\mathrm{grav}} h\, d_c^4}{128\, \eta\, \ell} \;\propto\; \frac{h}{\eta} \quad [\mathrm{kg/s}]. \tag{32}$$

PRL Fig. 1: $Q(h)$ linear at $H = 0$ ($\eta = 77\,\mathrm{mPa\,s}$) and at $B = 220\,\mathrm{mT}$ ($\eta = 220\,\mathrm{mPa\,s}$).

14.2 Data reduction, SI

$$\eta_r(H, f) = \frac{\eta(H, f) - \eta(0, 0)}{\eta(0, 0)}, \qquad \xi_0 = 0.0525\, B_0[\mathrm{mT}], \qquad \omega\tau_B = 10^{-2} f[\mathrm{Hz}]. \tag{33}$$

Gaussian (paper)	SI	ξ_0 / $\omega\tau_B$
2000 Oe	$159\,\mathrm{kA/m} = 200\,\mathrm{mT}$	$\xi_0 = 10.5$
2200 Oe	$175\,\mathrm{kA/m} = 220\,\mathrm{mT}$	$\xi_0 = 11.6$
800 Oe	$63.7\,\mathrm{kA/m} = 80\,\mathrm{mT}$	$\xi_0 = 4.2$
$f = 130\,\mathrm{Hz}$	—	$\omega\tau_B = 1.3$
$f = 700\,\mathrm{Hz}$	—	$\omega\tau_B = 7.0$

14.3 Findings

At $B \simeq 200\,\mathrm{mT}$ ($\xi_0 = 10.5$): $\eta = 220, 100, 77, 58\,\mathrm{mPa\,s}$ at $f = 0, 50, 250, 700\,\mathrm{Hz}$, i.e.

$$\Delta\eta: \; +143\,\mathrm{mPa\,s} \;\longrightarrow\; -19\,\mathrm{mPa\,s}, \qquad \frac{\eta_{\min}}{\eta(0,0)} = \frac{58}{77} = 0.75 \text{ (25\% reduction)}. \tag{34}$$

The $f = 0$ curve is the monotone McTague curve saturating at $\eta_r(\infty, 0) \simeq 2.5$; from $f \gtrsim 130\,\mathrm{Hz}$ ($\omega\tau_B \gtrsim 1.3$) η_r turns negative with deepening minima — the sign change at $\omega\tau_B \sim 1$, i.e. $f_0 \simeq 100\,\mathrm{Hz}$. Consistency: $\Delta\eta = -19\,\mathrm{mPa\,s} \Rightarrow g = (-19/77)/2.5 = -0.099$, matching the outermost experimental isoline -0.092.

14.4 Critical assessment (SI arithmetic)

1. τ_B **inconsistency.** Eq. (6) for a bare 10 nm grain: $\tau_B = 0.39\,\mu\mathrm{s}$ in water ($\eta = 1\,\mathrm{mPa\,s}$), $30\,\mu\mathrm{s}$ at $\eta = 77\,\mathrm{mPa\,s}$. Measured: 1.59 ms — a factor *52* above the latter. Inverting with solvent viscosity: $V_h = \tau_B k_B T/3\eta = 2.1 \times 10^{-21}\,\mathrm{m}^3$, i.e. $d_h \approx 159\,\mathrm{nm}$. The rotating units are aggregates/chains, while ξ_0 uses the single-grain m: a hybrid parametrization.
2. **Amplitude anomaly.** Dilute theory $\eta_r(\infty, 0) = \frac{3}{2}\varphi = 0.30$ vs. measured 2.5 (factor ~ 8): shape-only agreement after empirical normalization.
3. **Closure ambiguity** at intermediate ξ: up to 16% (Sec. 9).

15 Zero total viscosity and the pump instability

$\eta_{\text{total}} = 0$ requires

$$g = -\frac{2}{3\varphi} \overset{\varphi=0.2}{=} -\frac{10}{3}. \tag{35}$$

SM's critical point on the neutral curve: $\xi_c = 126$, $\omega_c \tau_B = 20$. In SI this demands

$$B_c = \frac{\xi_c k_B T}{m} = \frac{126 \times 3.99 \times 10^{-21}\,\text{J}}{2.09 \times 10^{-19}\,\text{A m}^2} = 2.4\,\text{T}, \qquad H_c = \frac{B_c}{\mu_0} = 1.9 \times 10^6\,\text{A/m}, \tag{36}$$

i.e. 2.4 T — well above the rigid-dipole ceiling $\mu_0 H_a = 1.0\,\text{T}$, so unattainable with these particles (larger ones ruin colloidal stability and Newtonian behaviour). States with $\eta_{\text{total}} < 0$ are *unstable states of rest* (spontaneous recirculation, "the system works as a pump"), not a steady material property. What is attained: continuous tuning $220 \to 50\,\text{mPa s}$ at fixed $B = 200\,\text{mT}$ by sweeping $f = 0 \to 700\,\text{Hz}$.

16 Gaussian → SI dictionary (summary)

Quantity	Gaussian	SI
Constitutive relation	$\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$	$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$
Dipole energy	$-\mathbf{m}\cdot\mathbf{H}$	$-\mu_0\mathbf{m}\cdot\mathbf{H} = -\mathbf{m}\cdot\mathbf{B}$
Langevin parameter	$\xi = mH/kT$	$\xi = \mu_0 mH/kT = mB/kT$
Susceptibility	$nm^2/3kT$	$\mu_0 nm^2/3kT$
Anisotropy field	$H_a = 2K/M_s$	$H_a = 2K/\mu_0 M_s$; $\mu_0 H_a = 2K/M_s$
Torque density	$\mathbf{M} \times \mathbf{H}$	$\mu_0\mathbf{M} \times \mathbf{H} = \mathbf{M} \times \mathbf{B}$
Kelvin force	$(\mathbf{M}\cdot\nabla)\mathbf{H}$	$\mu_0(\mathbf{M}\cdot\nabla)\mathbf{H}$
PRL (1)	$6\eta\varphi(\omega_p - \Omega) = \mathbf{M}\times\mathbf{H}$	$6\eta\varphi(\omega_p - \Omega) = \mu_0\mathbf{M}\times\mathbf{H}$
PRL (3) last term	$\frac{1}{2}\nabla \times (\mathbf{M} \times \mathbf{H})$	$\frac{\mu_0}{2}\nabla \times (\mathbf{M} \times \mathbf{H})$
PRL (4) last term	$-\frac{1}{6\eta\varphi}\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$	$-\frac{\mu_0}{6\eta\varphi}\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$
Key identity I	$\tau_B M_0 H = 3\eta\varphi\xi L$	$\mu_0\tau_B M_0 H = 3\eta\varphi\xi L$
Key identity II	$\tau_B \chi H_0^2 = \eta\varphi\xi_0^2$	$\mu_0\tau_B \chi H_0^2 = \eta\varphi\xi_0^2$
High-ω tail	$-\frac{\varphi}{\eta}(M_s\bar{H}/6\omega)^2$	$-\frac{\varphi}{\eta}(\mu_0 M_s\bar{H}/6\omega)^2$
τ_B, τ_s, ζ_r, I/τ_s	identical form (purely mechanical, no μ_0)	
FP eq., moment eq., PRL (5)–(9)	identical form (all μ_0 absorbed in ξ)	
$\Delta\eta = \frac{3}{2}\eta\varphi g$, g, R, SM (40)	identical (dimensionless)	

17 Verification ledger (SI-specific checks in bold)

Result	Verification performed
$\tau_B = 3\eta V/k_BT$	Einstein + $l(l{+}1)D_r$; **units** $\mathrm{Pa\,s\,m^3/J{=}s}$
$\xi = \mu_0 mH/k_BT$	**recomputed in SI:** 5.249×10^{-3} **per Oe vs. paper's** 5.25×10^{-3} **at** $T = 289\,\mathrm{K}$
$\chi = \mu_0 nm^2/3k_BT$	**dimensionless check** $\mathrm{H\,A^2/(m\,J){=}\,1}$**; value 1.76**
$H_a = 2K/\mu_0 M_s$	**gives** $7.96\times10^5\,\mathrm{A/m{=}10^4\,Oe}$**,** $\mu_0 H_a = 1.000\,\mathrm{T}$
$\langle\mathbf{ee}\rangle_0$, $L^2 + L' = 1 - 2L/\xi$	algebra; trace = 1; both limits; numerics to 10^{-6}
FP eq. (19)	Gibbs annihilation; norm; isotropy; μ_0 **correctly inside** U
Exact moment eq. (21)	operator identities; equilibrium stationarity exact
PRL (6)	coefficient-by-coefficient match; **form unchanged,** μ_0 **only in** $\boldsymbol{\xi}$
PRL (1),(3),(4) SI forms	**dimensional checks: torque density** = Pa**; last term of (4)** = A/(m s)
Identity I: $\mu_0\tau_B M_0 H = 3\eta\varphi\xi L$	**verified numerically to** 10^{-16} **with consistent** $\tau_B = 3\eta V/k_BT$
Identity II: $\mu_0\tau_B\chi H_0^2 = \eta\varphi\xi_0^2$	**verified numerically to** 10^{-16}
Identity III: $\mu_0\varphi M_s H_0\tau_B = 3\eta\varphi\xi_0$	**verified numerically to** 10^{-16}
$\Delta\eta = \frac{3}{2}\eta\varphi g$; PRL (2)	two proofs; **no** μ_0**, as required for a dimensionless ratio**
SM (2), MRSh static	full derivations; limits; tabulated difference
SM (28)	quasi-static factor check; ODE numerics at $\xi_0 = 0.3$ ($< 1\%$)
SM (29) SI form	$\bar{\xi}/\tau_B = \mu_0 M_s\bar{H}/3\eta$**; dimensional check to** Pa s
SM (40)	derived from scratch; reproduces 0.98, 1.707, −0.105 exactly
PRL (7),(8),(9)	line-by-line; static and weak-field limits; regular (ξ_e, P) form
PRL Fig. 3	independent RK4; Sec. 13 table
$\xi_c = 126 \Rightarrow B_c$	$B_c = \xi_c k_BT/m = 2.4\,\mathrm{T} > \mu_0 H_a = 1.0\,\mathrm{T}$
τ_B anomaly	**SI:** 0.39 µs **(water)** / 30 µs ($\eta = 77\,\mathrm{mPa\,s}$) **vs.** 1.59 ms**;** $d_h = 159\,\mathrm{nm}$

18 Errata (carried over and verified in SI)

1. PRL Eq. (3): printed $-\nabla\rho$; should be $-\nabla p$.
2. SM Eq. (40) as scanned, "$2 - \tanh\varepsilon - 2\tanh\varepsilon/\varepsilon$", is inconsistent with SM's own quoted numbers; correct form has $\tanh^2\varepsilon$ (Sec. 11).
3. Weak-field AC prefactor: $\Delta\eta = \frac{1}{8}\eta\varphi\xi_0^2 R$ (amplitude convention) $= \frac{1}{4}\eta\varphi\bar{\xi}^2 R$ (rms), with $g = \xi_0^2 R/12$; the pair $\frac{1}{4}\eta\varphi\xi_0^2$ *and* $g = \xi_0^2 R/12$ is internally inconsistent.
4. **SI-specific trap:** writing $\xi = mH/k_BT$ with H in A/m (i.e. forgetting μ_0) understates ξ by $\sim 10^6$ and destroys identities I–III. Equivalently, use $\xi = mB/k_BT$ with B in tesla.

19 Open problems for follow-up work

- Aggregation/polydispersity: average $g(\xi_0, \omega\tau_B)$ over a size distribution with $\tau_B(V) \propto V$, $m(V) \propto V$; does the hybrid parametrization emerge naturally?
- Beyond linear order in $\Omega\tau_B$: shear-rate dependence of g.
- General β; rotating/elliptic polarization within the MRSh closure.
- Exact Fokker–Planck benchmark of the MRSh closure in the driven regime (spherical-harmonic hierarchy) — in 2D this has been done and reveals sign errors of the closure at strong drive.
- Recompute the $\eta_{\mathrm{total}} = 0$ neutral curve with (PRL 6) instead of the Debye closure.